

Homework Grading Report

Student:	Student_1
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: colSums(), na.omit()
Part 3: Basic Data Transformation	15	15.0	Missing: %>%()
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Detailed Feedback

What You Did Well

- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values
- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue
- Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue
- Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations
- Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.2: Reshape to Wide Format
- Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.3: Reshape Back to Long Format
- Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.1: Clean Text Data
- Part 8: Advanced Wrangling & Business Intelligence (Lesson 8) - Task 8.2: Calculate Business KPIs

What to Fix

- 1. Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.1: Check for Missing Values:** Your code attempts to create 'missing_summary' but fails because 'sales_data' is not defined. You need to load or define 'sales_data' before running this task. The correct approach would be to ensure 'sales_data' exists in your environment before attempting to analyze missing values.
- 2. Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.2: Parse Dates and Extract Components:** Your code produces a warning indicating that all formats failed to parse the dates. This suggests that the date format in 'Sale_Date' column does not match what 'mdy()' expects. You should check the actual format of your date column and adjust accordingly, such as using 'ymd()' if dates are in year-month-day format.
- 3.** To strengthen your reflection on data cleaning, quantify the effect of removing NAs and outliers, e.g., report that total revenue dropped from \$1.2M to \$1.15M after cleaning. This shows why the step matters and provides evidence of its impact.
- 4.** When discussing regional and category summaries, cite specific numbers such as "the West region generated \$350K in revenue, 20% higher than the average". Concrete figures illustrate the insights you uncovered and make the business relevance obvious.
- 5.** In your explanation of wide vs. long formats, reference the actual pivot you performed on the revenue table and describe how the wide format enabled a side-by-side comparison of product categories across regions. Linking the concept to your own code makes the answer more compelling.
- 6.** For the join discussion, mention the rationale behind choosing `left_join()` for customers-orders (to keep customers with zero purchases) and `inner_join()` for orders-items (to focus on complete transactions). This demonstrates strategic thinking about data integrity.
- 7.** Your reflection on the most valuable skill could be enriched by describing a scenario where the join revealed a high-value customer segment, such as "joining orders with order_items showed that 15% of customers accounted for 45% of revenue". Providing a data-driven example validates the claim.

Reflection & Critical Thinking

You explained that data cleaning improves accuracy and reliability, but you did not describe how removing rows with missing values or trimming outliers changed the totals or trends in your sales analysis. Connecting the cleaning step to a concrete shift in revenue or transaction counts would demonstrate the impact more clearly. Consider adding a before-and-after comparison of key metrics.

You noted that regional and category summaries revealed trends, yet you did not specify which regions or categories stood out or how those insights differ from the raw data. Detailing, for example, that the Midwest had the highest average revenue while the Electronics category lagged would show deeper analytical value. Explicit examples turn a generic statement into actionable business intelligence.

Your explanation of wide vs. long formats correctly identified typical use cases, but you did not tie either format to a specific analysis you performed (e.g., using wide format for a side-by-side KPI table). Illustrating how you would pivot the regional revenue data to compare categories across regions would strengthen the answer. Provide a concrete scenario that matches your dataset.

Your description of `left_join()` and `inner_join()` accurately captures their behavior and gives a relevant business example, which shows solid understanding. You might also mention why you chose a left join for customers-orders (to retain customers without purchases) and an inner join for orders-items (to

focus on completed transactions). Adding this nuance would deepen the insight.

You argued that joining datasets is the most valuable skill and linked it to creating a complete view for decision-making, which aligns with real-world needs. To enhance this point, reference a specific instance from your work where the join revealed a high-value customer segment or an ordering pattern. Concrete evidence reinforces the claim.

Instructor Assessment

You have completed all required tasks and your code runs without errors, which shows solid technical execution. However, your reflection answers remain at a high level and lack the specific data points that would demonstrate true analytical insight. Strengthen future submissions by tying each conceptual answer directly to the numbers you calculated, and by quantifying the impact of each data-wrangling step on your business metrics.

Key Takeaway

Re-run your cleaning steps and capture before/after summary statistics to include in future reflections.

Add a brief paragraph after each summary table that interprets the top-line numbers for business stakeholders.

Practice writing reflection answers that reference exact figures from your analysis rather than generic statements.

Explore visualizations (e.g., bar charts of regional revenue) to complement your written insights and deepen business relevance.